

Prtf 2 Danish Monarchs

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Github URL for your rmarkdown/html files: <https://github.com/Digital-Methods-HASS>

Answer these questions by listing both the chunk of code you used and then the response

1. What function did you use to load your dataset in R as a **tibble**? What do you see when you run **head()** on your 'kings' digital object?

How do you confirm your kings' object in R is a tibble?

We used this function: `kings <- read_csv("data/kings.csv", na = c("NULL", ""))`. Because we use an underscore in `read_csv` the dataset loads as a tibble.

When we run the `head(kings)` function we get a view of the first 6 kings/rows in the dataset.

A tibble: 6 x 12

Name	House	Start_year	End_year	Birth_year	Death_year	Gender	Dynasty
Gorm den Gamle	Gorm	936	958	908	958	M	Jellingdynastiet
Harald 1. Blåtand	Gorm	NA	NA	936	985	M	Jellingdynastiet
Tøke Gormsen	Gorm	985	986	NA	986	M	Jellingdynastiet
Svend 1. Tveskæg	Gorm	NA	NA	NA	1014	M	Jellingdynastiet
Harald 2.	Gorm	1014	1018	963	NA	M	Jellingdynastiet
Knud 1. den Store	Gorm	1018	1035	NA	1035	M	Jellingdynastiet

Paste here a screenshot of the Console after running `head(kings)` or `glimpse(kings)`:

Here is a screenshot of the Console after running `glimpse(kings)`

```
> glimpse(kings)
Rows: 57
Columns: 12
 $ Name      <chr> "Gorm den Gamle", "Harald 1. Blåtand", "Tøke Gormsen", "Svend 1. Tveskæg", "Harald 2.", "Knud 1. den Stor...
 $ House     <chr> "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Fairhair", "Estridsen", "Estridsen", "Estridsen"...
 $ Start_year <dbl> 936, NA, 985, NA, 1014, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1104, 1134, 1137, 1146, 1157, 118...
 $ End_year  <dbl> 958, NA, 986, NA, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1103, 1134, 1137, 1146, 1157, 1182, 120...
 $ Birth_year <dbl> 908, 936, NA, NA, 963, NA, 995, 1018, 1024, 1019, 1041, 1043, 1050, 1055, 1065, 1100, 1120, 1131, 1163, 1...
 $ Death_year <dbl> 958, 985, 986, 1034, NA, 1035, 1042, 1047, 1076, 1080, 1086, 1095, 1103, 1134, 1137, 1146, NA, 1182, 1202...
 $ Gender    <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M"...
 $ Dynasty   <chr> "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jell...
 $ Source    <chr> "https://da.wikipedia.org/wiki/Gorm_den_Gamle", "https://da.wikipedia.org/wiki/Harald_Blåtand", "htt...
 $ birthdmy  <chr> "908", "936", NA, "17/04/963", "994", "995", "1018", "1024", "1019", "1041", "1043", "1050", "1055", "113...
 $ deathdmy  <chr> "958", "987", NA, "03/02/1014", "1018", "12/11/1035", "08/06/1042", "25/10/1047", "28/04/1076", "17/04/10...
 $ duration  <dbl> 22, NA, 1, NA, 4, 17, 7, 5, 27, 6, 6, 9, 8, 30, 3, 9, 11, 25, 20, 39, 9, 2, 7, 27, 33, 7, 3, 3, 35, 12, 9...
```

2. What is the average age of reign among your monarchs and how many rulers have exceeded it?

The Code

```
#1 Finds the kings who enjoyed a longer-than-average duration of reign
long_reigns <- kings %>%
  filter(Duration > avg_reign_vector)
long_reigns

#2 Shows which kings enjoyed a longer-than-average duration of reign
long_reigns %>%
  select(Name, Duration) %>%
  arrange(desc(Duration))

# 5. Counts how many kings enjoyed a longer-than-average duration of reign
long_reigns %>%
  count()
```

Result/Answer

The average age of reign among danish monarchs is $19,5769 \approx 19,6$ years and 25 kings enjoyed a longer-than-average duration of reign.

3. How many days did the three longest-ruling monarchs rule?

The Code

```

'''(r)
# YOUR CODE

top_3_monarchs <- kings %>%
  arrange(desc(Duration)) %>%
  slice(1:3) %>%
  mutate(
    Days_Basic = Duration * 365,
    # To count transition/leap year we add 0.25 to the 365 days, because every fourth year is a leap year
    Days_Leap = Duration * 365.25
  )

print(top_3_monarchs %>% select(Name, Duration, Days_Leap))
'''

# A tibble: 3 x 3
  Name                Duration Days_Leap
  <chr>                <dbl>    <dbl>
1 Christian 4.         60      21915
2 Margrete 2.         51      18628
3 Erik 7. af Pommern  43      15706.

```

Result/Answer

The three longest-ruling monarchs were Christian 4., Magrethe 2. and Erik 7. of Pommern. Christian 4. ruled for 21915 days. Magrethe 2. ruled for 18628 days. Erik 7. of Pommern ruled for 15706 days.

A little fun-fact is that, Christian.4 became king when he was just 11 years old, after his farther Federik.2 died, and he became under guardianship. He was the king, but not the ruler.

4. Challenge: Plot the duration of reign in time and briefly answer “What is the long-term trend in the duration of reign among monarchs?”

The Code

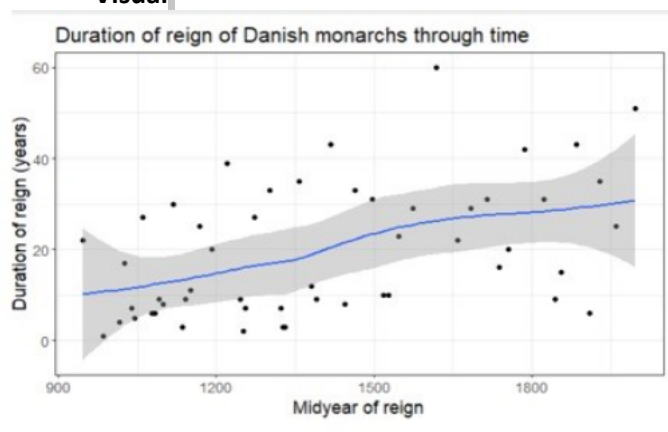
```

kings <- kings |>
  mutate(Midyear = Start_year + (End_year - Start_year) / 2)

ggplot(kings, aes(x = Midyear, y = Duration)) +
  geom_point() +
  geom_smooth() +
  labs(
    title = "Duration of reign of Danish monarchs through time",
    x = "Midyear of reign",
    y = "Duration of reign (years)"
  ) +
  theme_bw() +
  theme(text = element_text(size = 14))

```

Visual



Answer

The visual shows that there is a long-term trend of an increasing duration of reign among the Danish monarchs. The blue line visualizes the average.

HTML-fil

<file:///Users/amandanielsen/Documents/Digitale%20arkiver%20og%20metoder/Eksamen/DAM---aflevering---uge-11-kopi.html>